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Endogeneity

1. Definition

Endogeneity happens when an independent variable is correlated with the disturbance term (i.e., error term) in regression modeling (Lam et al., 2016). And the consideration of endogeneity should be based on substantive theoretical considerations (Ketokivi & McIntosh, 2017). If the endogeneity issue is ignored, the biased estimation results can be obtained, leading to distort causal relationship, and thus the incorrect conclusions and managerial insights.

2. Sources of Endogeneity

There are three sources of endogeneity: (1) omission of variables (2) errors-in-variables (3) simultaneous causality (Wooldridge, 2002). For panel data, the sources of endogeneity include unobserved heterogeneity, simultaneity and dynamic endogeneity (Wintoki et al., 2012; Yiu et al., 2020). Dynamic endogeneity occurs when the current values of a study's independent variables are affected by the past values of the dependent variables (Yiu et al., 2020).

2.1. Omission of Variables

The omission of variables in regression models can cause endogeneity. When the omitted variable is correlated with the dependent variable and the independent variable, then the independent variable should be associated with the disturbance term since all omitted and unobservable variables are included in the disturbance term. Under this situation, the assumption of exogeneity is violated and the endogeneity occurs (Kennedy, 2008; Wooldridge, 2002).

And the omission of variables can be caused by the unavailability of data. For example, Wiengarten et al. (2019) only use sample firms with complete data, by removing sample with missing date in the sample period. To ensure no selection bias in the sample will bias their result, the independent t-tests are conducted between the discarded firms and retained firms to examine whether the missing data are random or systematic.

Omitting selection is a conventional type of the endogeneity caused by omission of variable (Qiu et al., 2022). Sample self-selection based on observables can be controlled by using control variables. Unfortunately, the sample self-selection based on unobserved variables that are correlated with the dependent and independent variables can cause the endogenous problems. This situation can result in the observed values of dependent variables of the sample are different from non-sample parts on a nonrandom basis (Fan et al., 2022; Qiu et al., 2022; Yeung et al., 2011).

2.2. Errors-In-Variables

Errors-in-variables is also a source that can cause endogeneity. The endogeneity occurs when variables are inappropriately measured and their true values are still unobserved (Wooldridge, 2002). Measurement errors occurs when the constructs in the theoretical model are not measured perfectly, or non-comprehensiveness of the data collection method since missing data is also deemed as a type of measurement error (Kennedy, 2008). Besides, if the variables that the researchers collect data for differ from the variables that truly affect the phenomena to be studied, the errors-in-variable will also negatively influence the results of the research (Wooldridge, 2002).

Measurement error in dependent variables will not cause endogeneity. However, measurement errors in independent variables may cause severe endogenous issues depend on the assumptions about the measurement error (Wooldridge, 2002). The first assumption is that the measurement error and the observed independent variable are uncorrelated, and that the error term of the model is uncorrelated with the actual (unobserved) and the observed independent variable. In this case, estimation yields consistent coefficients. The second assumption is also known as the ‘classical errors-in-variable (CEV) assumption’. Under this assumption, the measurement error is not correlated with the actual (unobserved) independent variable, and the disturbance term in the regression model is also uncorrelated with the actual (observed) independent variable. Under this situation, the observed independent variable is correlated with the measurement error, and thus it is correlated with the disturbance term, violating the fundamental assumption for OLS model and resulting in endogeneity. In other words, the inconsistent estimation of the coefficient, which is biased towards zero, will be obtained from the regression model. And the size of this bias depends on the variance of the actual independent variable relative to the variance in the measurement error (Ketokivi & McIntosh, 2017).

2.3. Simultaneous Causality

Simultaneous causality, also known as reverse causality, can also cause endogeneity, when one (or more) independent variable is jointly determined with the dependent variable (Wooldridge, 2002). In other words, the independent variables and dependent variables cause each other at the same time and the causal effects for them are reciprocal and independent and dependent variables at the same time cause each other (Wooldridge, 2002).

3. Solutions to Endogeneity

Because the disturbance term in endogeneity issue is unobservable, there is no direct method to statistically test whether the endogenous variable is correlated with the disturbance term. Therefore, only indirect tests and precautionary measures can assist the research to make the decision of how to mitigate endogeneity in regression modeling.

The endogeneity issue can be mitigated at two stages, including the research design stage and the statistical analysis stage. At the research design stage, researchers can extensively reviewing literature and providing comprehensive research designs that could help to apply appropriate statistical tools (Ketokivi & McIntosh, 2017). As for the statistical analysis, research should ensure the data used in the research is rigorously invested by statistical techniques (Ketokivi & McIntosh, 2017). Besides the research design, there are additional factors that need to be considered when deciding the appropriate method to mitigate endogeneity in research, such as data collection instrument, sample size, complexity of the model, and underlying theory and research context. In this report, the solutions to endogeneity will be summarized and stated for the two stages, i.e., the research design stage and the statistical analysis stage.

3.1. Solutions at Research Design Stage

Research design is a strategy for answering the research question using empirical data. And the research design includes the following decisions:

- (a) Overall research objectives and research approach
- (b) Type of research: primary research or secondary research
- (c) Sampling method

(d) Data collection methods

(e) Data analysis methods

To address the endogeneity issue in OM empirical research, theoretical, empirical and pragmatic considerations should be made. And the solutions that can be used to mitigate endogeneity at the research design stage are summarized and analyzed below.

3.1.1. Theoretical Prescription

There are some theoretical steps for researchers to mitigate the endogenous issues in their research. First, the research should list all the potential endogeneity concerns implied by the theoretical model, and decide how significant a threat each of the listed concerns generate to empirical analysis (Ketokivi & McIntosh, 2017). This procedure require a thorough understanding of the theoretical model. Researchers should understand the structure of the model and to interpret the assumptions embedded in the structural limitations of the model, and explore how and why the relevant factors of interest are interrelated (Ketokivi & McIntosh, 2017). Second, the research should identify potential omitted variables at the model specification stage (Ketokivi & McIntosh, 2017).

All confounding events that can influence the independent variables and the dependent variable should be identified as much as possible based on theory and extant literature. Researchers should make judgement about whether there is any omitted variable exist form the theoretical aspect when building the theoretical model for the research. However, there are confounding events that cannot be identified, such as the confounding events on unobservable variables (Fan et al., 2022; Qiu et al., 2022; Wiengarten et al., 2019; Yiu et al., 2020) and the sample selection bias based on unobservable factors (H. Li et al., 2022; Qiu et al., 2022; Wiengarten et al., 2019). For example, in the research, examing irresponsible behavior by exploring the role of operations management in violating or breaching environmental and safety regulations, a matched-sample design are used, where each matched pair needs to be controlled for due to the fixed effects related to the year of the breach as well as the industry context, which includes regulation, macroeconomic conditions, and industry seasonality (Lo et al., 2022). Under such situation, the remedies for endogeneity problems should be considered in order to make causal inference.

3.1.2. Empirical Remedy

After the theoretical and substantive concerns be handled, empirical remedies can be considered by researchers to solve the endogeneity. In this report, some common approaches are discussed.

3.1.2.1. Experimental Research

3.1.2.1.1. Definition

In experimental research, researchers systematically intervene in a process and measure the outcome. And to carry out an experiment, researchers should vary the independent variable, accurately measrue the dependent variable, and and control for confounding variables. This method can generate more clear casual inference than other non-experiment research method. The validity of research depends on the experimental design and different experimental design has different level of internal and external validity. There are four diferent types of experiments that are laboratory experiment, scenariobased experiment, natural experiment, and quasi-experiment (Gao et al., 2023). A laboratory (or lab) experiment is an experiment carried out inside of laboratory settings, in which researchers have precise control over various conditions, including

where the experiment takes place, when to implement, whom to include as participants, and what interventions to use (Gao et al., 2023). A vignette or scenario-based experiment is an experiment in which varying versions of a descriptive vignette are deployed to convey scripted information about specific levels of factors of interest (i.e., independent variables) to human subjects (Gao et al., 2023). A natural experiment is an observational study in which individuals are exposed to experimental and control conditions that are determined by factors outside the control of investigators (Gao et al., 2023). That is, natural experiments occur in real-life environments where individuals are exposed to treatment or control conditions, rather than being designed and implemented by researchers. A quasi-experiment is an interventional study used to estimate the causal impact of an intervention on the target population without random assignment (Gao et al., 2023). The laboratory experiment (Carter & Stevens, 2007), the natural experiment designs (Fan et al., 2022; Lam et al., 2022) and can be used to solve the endogeneity issues. The quasi-experiment design itself does not solve endogeneity given the non-randomly assigned intervention can introduce the selection bias problems. The difference-in-difference (DID) approach is a quasi-experiment design commonly used to solve the endogeneity in OM empirical research (Fan et al., 2022; Lam et al., 2022; Miao et al., 2023). And the DID approach and matching approaches are discussed at the section “Longitudinal, quasi-experimental data”. Therefore, only the natural experiment design and regression discontinuity design (RDD) are discussed in this part.

3.1.2.2.2. Natural Experiment Design

A natural experiment is an observational study in which individuals are exposed to experimental and control conditions that are determined by factors outside the control of investigators (Gao et al., 2023). The main difference between the natural experiment and the quasi-experiment is that the criterion for assignment is selected by the researcher in the quasi-experiment while the assignment occurs 'naturally,' without the researcher's intervention in a natural experiment. Natural experiments occur in real-life environments where individuals are exposed to treatment or control conditions, rather than being designed and implemented by researchers. Therefore, the treatment in natural experiment design is “as-if random” because there is no explicit randomization process designed by researchers. Therefore, the natural experiment design can solve the endogeneity caused by selection bias.

In the research in OM empirical field, a natural experiment to understand the influence of treatment (or intervention) by comparing the observations from the treatment group and control group. Propensity score matching (PSM) is used to develop the matched pairs. The matching procedures ensured the treatment and control firms were highly similar in terms characteristics (confounding factors) other than the variables of interest. And then, DID test is performed to make causal inference (Adbi et al., 2019; Cui et al., 2022; Fan et al., 2022). The details of DID test and PSM will be discussed in the corresponding parts of them. Other research combine the an event analysis methodology with the natural experiment design (Pan et al., 2020).

3.1.2.2.2.1. Advantages of Natural Experiment Design

There are several advantages of the natural experiment design. First, participants are unlikely to anticipate or be aware of being observed, thereby minimizing the concern of the Hawthorne effect. Second, the real-life environment also engenders high ecological validity of the findings and reduces participants' likelihood of not complying with the experimental conditions. Third, the

natural setting is extremely helpful in situations where the randomization of experimental conditions is ethically unacceptable (Gao et al., 2023).

3.1.2.2.2. Disadvantages of Natural Experiment Design

Unfortunately, there are many problems to perform an experimental research in the operations management research. First, if the experiments are carried out in the real-life environment rather than in the laboratory, the researchers may not be able to manipulate the interested variables and examine the related effects. And the absence of researcher control over the experiment poses challenges to causal inference and replication of the study (Gao et al., 2023). Second, it is often a challenging task to identify a natural shock in the first place (Gao et al., 2023).

3.1.2.2.3. Regression Discontinuity Design

Regression Discontinuity Design (RDD) is a quasi-experimental evaluation option that measures the impact of an intervention, or treatment, by applying a treatment assignment mechanism based on a continuous eligibility index which is a variable with a continuous distribution. It is a method to deal with non-random treatment effects. The major idea of this method is to find a factor that can describe how an observation becomes part of the treatment group and seeks to exploit the cut-off point for this identified factor (Calvo et al., 2019; T. Chen et al., 2022). The discontinuity in this method refers to identifying the threshold or the cut-off point that can distinguish the treatment from the control group (Calvo et al., 2019; T. Chen et al., 2022). RDD can solve the endogeneity issues caused by selection bias and it can deal with non-observable characteristics more convincingly than other quasi-experimental matching methods

There are two different RDD methods that are fuzzy RDD (Cao et al., 2019; T. Chen et al., 2022; Cheng et al., 2020) and sharp RDD (Calvo et al., 2019; Flammer, 2015). In fuzzy RDD, the cut-off point increases the probability of treatment but doesn't completely determine treatment and the probability of treatment is discontinuous at the cutoff, but not to the degree of a definitive 0 to 1 jump (Cao et al., 2019; T. Chen et al., 2022; Cheng et al., 2020). In sharp RDD, the assignment variable completely determines treatment and the probability of treatment changes from 0 to 1 at the cut-off (Calvo et al., 2019; Flammer, 2015).

In RDD, the assignment variable should be a continuous characteristic variable (Calvo et al., 2019). Besides, to effectively apply RDD, the assumption that the value of the assignment variable around the cut-off point is randomly distributed, which means that whether the value of the assignment variable is larger or smaller than the threshold is as good as randomly assigned (Cao et al., 2019; T. Chen et al., 2022; Flammer, 2015). Meanwhile, the choice of the cut-off point should not be influenced by the assignment variable (Calvo et al., 2019). And except for the treatment status that jumped at the cut-off point, there should be no significant difference in other individual non-treated characteristics at the cut-off point. Otherwise, the changes in observation results at the cut-off point can be caused by the other characteristics than the treatment (Cao et al., 2019; T. Chen et al., 2022; Cheng et al., 2020). And in RDD, there are two strategies for estimating the discontinuity: a parametric (global) strategy and a nonparametric (local) strategy (Calvo et al., 2019).

3.1.2.2.3.1. Advantages of RDD

There are several advantages of RDD. First, RDD is a approach to deal with non-random treatment effects (Cao et al., 2019; T. Chen et al., 2022; Flammer, 2015). Second, RDD can solve the

endogeneity issues caused by selection bias. RDD deals with non-observable characteristics more convincingly than other quasi-experimental matching methods. This method allows researchers to compare subjects that are just above the cutoff point against those subjects that are marginally below the cut-off point (Calvo et al., 2019; T. Chen et al., 2022). When properly implemented and analysed, the RDD yields an unbiased estimate of the local treatment effect. The RDD can be almost as good as a randomised experiment in measuring a treatment effect (Calvo et al., 2019; Flammer, 2015). And RDD, as a quasi-experiment, does not require ex-ante randomisation and circumvents ethical issues of random assignment (Calvo et al., 2019; Flammer, 2015). Meanwhile, well-executed RDD studies can generate treatment effect estimates similar to estimates from randomised studies.

3.1.2.2.3.2. Disadvantages of RDD

There are also many disadvantages of RDD. First, the estimated effects are only unbiased if the functional form of the relationship between the treatment and outcome is correctly modelled. The most popular caveats are non-linear relationships that are mistaken as a discontinuity (Cao et al., 2019; T. Chen et al., 2022; Cheng et al., 2020). Second, the contamination by other treatments may exist (Calvo et al., 2019; Flammer, 2015). Suppose another treatment occurs at the same cutoff value of the same assignment variable. In that case, the measured discontinuity in the outcome variable may be partially attributed to this other treatment. Third, the selection criteria and/or threshold are not always clear and the sample may be insufficiently large for the analysis. In addition, RDD yields a ‘local area treatment effect’. That is, the impact estimate is valid for those close to the threshold, but the impact on those further from the threshold may be different (it could be more or less). In practice, however, where it has been possible to compare this ‘local’ effect with the ‘average’ effect, the differences have not been great. This indicates that RDD is an acceptable method for estimating the effects of an intervention, such as the policy.

3.1.2.2. Instrumental Variables Techniques

The instrumental variables (IVs) are variables that are uncorrelated with the disturbance but are correlated with the endogenous independent variable in the original regression model (Murray, 2006). And IVs do not influence the dependent variable directly, instead they influence it indirectly through the endogenous independent variables. The fundamental idea of IVs is to decompose the variations in the endogenous independent variable with the help of IVs by concentrating on the variations in the endogenous independent variable that are uncorrelated with the disturbance term in the model and disregarding the variations that bias the estimation (Wiengarten et al., 2017). The instrumental variables techniques can mitigate there types of endogeneity, including omitted-variable endogeneity, endogeneity caused by errors-in-variables and endogeneity caused by simultaneous causality. Unlike matching method that can only solve problems due to the selection on observables, IVs can solve problems due to the selection on unobservables, which arises when a regressor (in our case, a treatment) correlates with the disturbance term (Mithas et al., 2022).

To effectively use IVs to mitigate endogeneity, valid IVs should be identified. There are two conditions based on which the validity of IVs can be judged. These two conditions are relevance and exogeneity (Kennedy, 2008; Murray, 2006; Wiengarten et al., 2017; Yiu et al., 2020). Relevance refers to the degree to which an instrumental variable is sufficiently correlated with the endogenous independent variable. And an instrumental variable that has a high correlation with the endogenous independent variable is defined as the strong instrumental variable. In contrast, an

instrumental variable that has a low correlation with the endogenous independent variable is defined as the weak instrumental variable. Exogeneity is defined as the degree to which an instrumental variable and the disturbance term in the model are uncorrelated (Murray, 2006).

3.1.2.2.1. Two Stage Least Square (2SLS)

2SLS is used for over-identified models. And it can be used if the disturbance term is homoscedastic or heteroscedastic and/or auto-correlated (Qiu et al., 2022). First, it uses IV to compute the estimated values of the suspected endogenous variables, which means an OLS-regression on the instrument (the first stage). Second, it uses those computed values to estimate a linear regression model of the dependent variable, which means the OLS-regression of the fitted values from the first stage regression (the second stage). Since the computed values are based on variables that are uncorrelated with the errors, the results of the two-stage model are optimal. The detailed two steps are shown as follow:

Step 1: Regress the endogenous variable on all chosen instruments, which have previously undergone relevance and exogeneity checks, and obtain the residual for the endogenous variable.
Step 2: Replace the endogenous variable with the corresponding residual and regress the dependent variable on it.

3.1.2.2.2. Advantages of IVs

This method has many advantages. IVs method allows for the possibility of making causal inferences with observational data and it can adjust for both observed and unobserved confounding effects (Qiu et al., 2022). It is one of the most commonly used method to solve endogeneity in empirical research due to its simplicity and effectiveness. And it can simultaneously deal with the endogeneity problems caused by several different sources includes the omitted-variables (including observable and unobservable variables), selection bias, measurement errors-in-variables and simultaneous causality (Qiu et al., 2022; Yiu et al., 2020). And for overidentifying situations where the number of IVs are larger than that of the suspected endogenous variables, the test on the exogeneity of the instruments is possible.

3.1.2.2.3. Disadvantages of IVs

There are many disadvantages of IVs to be used in OM empirical research to solve endogeneity. First, it is difficult or even impossible to obtain strictly exogenous instruments externally (Roodman, 2009; Wiengarten et al., 2017; Wintoki et al., 2012; Yiu et al., 2020), i.e., the valid instrumental variables that are correlated with the endogeneous variable but not correlated with the disturbance. This difficult is particularly evident in the research with a large number of endogenous variables included in the research and the limited availability of appropriate external datasets that cover the samples of the research (Yiu et al., 2020). The incorporation of disqualified instrumental variable may not solve the endogenous issues but make the estimations obtained from the regression model become even worse. Second, the precision of instrumental variable regression estimates is lower than that of OLS estimates (Baum et al., 2003). Although the use of instrumental variable can mitigate estimation inconsistency, however, the estimation will become less accurate. If only weak instruments are available, the loss of precision will be severe and instrumental variable regression estimates may have no improvement over OLS estimates. This efficiency loss can be controlled by using a large sample. Third, most of the justification for the use of instrumental variable regression is asymptotic and the regression technique may not work well in

small samples. Using instrumental variables can cause the increase of the finite-sample bias. Finally, the instrumental variable techniques are based on the assumption that at least one of the instrumental variable is exogenous. This means that at least some of them are untestable. And there is no way to ensure that this assumption definitely holds in research.

3.1.2.3. Examine Variance and Change Over Time

Panel data contains multiply subjects. It is a subset of longitudinal data where observations are for the same subjects each time. And two relevant variants of longitudinal data: observational and quasi-experimental data.

3.1.2.3.1. Longitudinal, Observational Data

Panel data contains multiply subjects. It is a dataset in which the behavior of each subject is observed at multiple points in time. Thus, it is a subset of longitudinal data. There are two different types of model in panel data, i.e. the static panel data model and the dynamic panel model. There are two different types of model in panel data, i.e. the static panel data model and the dynamic panel model. The static panel data model includes fixed effects (FE) model and random effects (RE) model, which can mitigate endogeneity caused by omitted variables, including unobserved individual heterogeneity and time heterogeneity (Hsiao, 2022). And the dynamic panel model includes the least squares dummy variables (LSDV) estimator, which can solve the endogeneity caused by omitted variables, including serial correlation of the dependent variable. (Lam et al., 2016; Yiu et al., 2020). Meanwhile, the generalized method of moments (GMM) method for dynamic panel data is also discussed in this part and GMM method can mitigate endogeneity caused by omitted variable, errors-in-variables and simultaneous causality.

3.1.2.3.1.1. Static Panel Data Model

The static panel data model does not allow to use lagged values of the dependent variable. And the two static panel data models, i.e. the static panel data and the dynamic panel data model, are discussed in details below.

3.1.2.3.1.1.1. Random Effects Model

To study the time-invariant effects, a model includes time-invariant regressors should be used. And this is called random effects (RE) model. The model of RE is showed as follow:

$$Y_{it} = \alpha + \beta x_{it} + \gamma z_i + e_{it}$$

Where the disturbance term $e_{it} = \alpha_i + u_{it}$, and α_i is an unobservable and time-invariant variable that is correlated with the dependent variable Y_{it} . x_{it} and z_i are independent variables (i.e., regressors), where x_{it} is an observable variable that varies over the time and z_i is an observable time-invariant variable. The key assumption of RE is that the omitted variable α_i is uncorrelated with the included independent variables x_{it} and z_i . This method can mitigate the endogeneity caused by omitted variables (Bolton & Katok, 2008; Ibrahim & L'Ecuyer, 2013).

3.1.2.3.1.1.1.1. When to Use Random Effects Model

If the individual effects are strictly uncorrelated with the independent variables, it may be suitable to use the random distribution across cross-sectional units to model the individual specific constant terms, i.e., α_i in the regression model. This is appropriate if the that sampled cross-sectional units are drawn from a large population (Bolton & Katok, 2008).

If the differences across subjects have influence on the dependent variable of theoretical interest, the RE should be included into the model. In a RE model, researchers should clarify whether those individual characteristics, such as α_i in the regression model, may or may not influence the independent variables.

3.1.2.3.1.1.2. Advantages and Disadvantages

The advantage of using RE method is that time invariant variables (e.g., geographical contiguity, distance between states) can be included in the regression model. In the fixed effects model, these variables are absorbed by the intercept (Hsiao, 2022).

There are also some disadvantages of RE models. First, the challenge with the application of RE is that RE must be assumed to be exogenous to all of the predictor variables, which may not be possible in research due to realistic restrictions, such as data is not available (Hsiao, 2022). Second, another concern of RE method is that data on some variables (i.e., individual characteristics such as innate ability) may not be available, hence leading to omitted variable bias in the model.

3.1.2.3.1.1.2. Fixed Effects Model

Fixed effects method employs panel data to control for variables that differ across subjects, but are constant over time. Because adding the FE can alleviate unobserved heterogeneity due to omitted variables, this method can mitigate endogeneity caused by omission of variable. There are two fundamental assumptions of FE. The first assumption is that the omitted variable α_i is correlated with the independent variables. The second assumption is that the unobserved variable α_i is correlated with independent variables but does not differ over time.

The model of FE is showed as follow:

$$Y_{it} = \alpha + \beta x_{it} + \gamma z_i + \alpha_i + u_{it}$$

Where u_{it} is the disturbance term. In the FE model, the unobservable variables are correlated with the independent variables, therefore, the individual effect α_i should not be included in the disturbance term, instead, it should be treat as an explanatory variable. In this way, the FE model control for unmodeled differences that are both stable over time and that may correlate with the predictors of interest. In other words, this model remove all the unmodeled time-invariant effects from the dependent variable, therefore, researchers can assess the net effect of independent variables on the dependent variable. In fixed effects models, the slope coefficient of the population regression line is the same for all subjects, but the intercept of the population regression line varies across subjects (Narayanan & Terris, 2020).

3.1.2.3.1.1.2.1. Estimation

The report discusses two different methods to estimate fixed effects models: (i) within estimator (ii) dummy variable estimator.

3.1.2.3.1.1.2.1.1. Within Estimator

This is the more commonly used estimator for FE models. This estimator is called the "within estimator", as it uses time variation within each cross-section (Ibrahim & L'Ecuyer, 2013).

3.1.2.3.1.1.2.1.2. Dummy Variable Regression

When there are a small number of fixed effects to be estimated, it is convenient to just run dummy variable regression for a FE model (Bolton & Katok, 2008; Narayanan & Terris, 2020).

3.1.2.3.1.2.1.3. Two-Way Fixed Effects Model

Two-way fixed effects model is a linear regression with unit and time fixed effects (Lu et al., 2018). Many OM empirical research control time fixed effects and individual fixed effects (Jiang et al., 2023; Yiu et al., 2020).

3.1.2.3.1.2.1.4 Advantage and Disadvantage of FE Models

As for the advantage, if researchers only aim to control for the effect of a time-invariant variable, such as industry or country of incorporation, FE will be an effective and appropriate model (Adbi et al., 2022; Bolton & Katok, 2008; Ibrahim & L'Ecuyer, 2013; Jiang et al., 2023; Narayanan & Terris, 2020).

However, there are some disadvantages of FE method. First, FE do not work when lagged outcomes are included in the regression. Therefore, the lagged dependent variable should not be used as a regressor in FE model. Second, the FE model therefore cannot include any predictors that are stable over time (Ketokivi & McIntosh, 2017). Because any independent variable of theoretical interest that is stable over time will be highly collinear with the FE, which results in estimation problems.

3.1.2.3.1.3. Fixed Effects or Random Effects?

The Hausman test to decide whether to use a fixed effects or random effects model. Procedures are: (1) Run a fixed effects model and save the estimates (2) Run a random effects model and save the estimates (3) Perform the Hausman test.

Both FE and RE models both assume all regressors-including the unobserved effect-to be uncorrelated with the disturbances of the dependent variable. The choice of RE versus FE thus constitutes a tradeoff: FE relaxes the assumption of no correlation between the unobserved effect and the predictor variables, but simultaneously, preempts the modeling of time-invariant predictors (Ketokivi & McIntosh, 2017).

3.1.2.3.1.2. Dynamic Panel Data Model

In dynamic panel data models, dummy variables may be introduced to the least squares to explain the effect of each individual unit of a cross section which is unobserved but correctly specifies the model of relation, which is known as the least squares dummy variables (LSDV) estimator (Kiviet, 1995). In OM empirical, problems can be caused by the serial correlation of the dependent variable, when making causal inference. For example, the realization of the current firm performance is correlated with its past performance (Lam et al., 2016). To solve this problem, researchers include lagged dependent variables in the models to account for the persistent influence of past performance. However, the lagged dependent variables are correlated with the fixed effects in the error term by construction, leading to dynamic panel bias (also known as Nickell bias) (Yiu et al., 2020). LSDV estimator is able to handle the dynamic panel bias, it works only for balanced panels (Roodman, 2009) and is unsuitable for the unbalanced sample with some samples having more observations than the others (Lam et al., 2016).

3.1.2.3.1.2.1. Advantages and Disadvantages of LSDV

The LSDV estimator has several advantages. First, the LSDV estimator can solve the endogeneity caused omitted variables, including the serial correlation of the dependent variable (Lam et al., 2016; Yiu et al., 2020). Second, the LSDV estimator can solve the dynamic bias or the Nickell bias (Lam et al., 2016; Yiu et al., 2020).

There are also many disadvantages of the LSDV estimator. First, the LSDV estimator is able to handle the dynamic panel bias, it works only for balanced panels (Roodman, 2009) and is unsuitable for our unbalanced sample with some firms having more observations than the others (Lam et al., 2016). Second, the LSDV estimator can solve the dynamic bias or Nickell bias (Kiviet, 1995), but it does not deal with the other endogeneity concerns such as unobservable heterogeneity and simultaneity arising from the other regressors included in the research (Roodman, 2009; Yiu et al., 2020).

3.1.2.3.1.3. Advantage and Disadvantage of Panel Data

Panel data, by blending the inter-individual differences and intra-individual dynamics, have several advantages over cross-sectional or time-series data (Hsiao, 2022). First, panel data can provide more accurate inference of model parameters. Panel data usually contain more degrees of freedom and more sample variability than cross-sectional data which may be viewed as a panel with $T = 1$, or time series data which is a panel with $N = 1$, hence improving the efficiency of econometric estimate (Hsiao, 2022). Second, It has greater capacity for constructing more realistic behavioral hypotheses. By blending inter-individual differences with intra-individual dynamics, longitudinal data allow a researcher to analyze a number of important economic questions that cannot be addressed using cross-sectional or time series data sets alone (Hsiao, 2022). Third, It can uncover dynamic relationships based on the dynamic panel data model. Forth, panel data can mitigate endogeneity caused by omitted variable, errors-in-variables and simultaneous causality.

For the disadvantage, the panel data may not available in research. The panel datasets combine the characteristics of cross-sectional and time series data. Although, the emergence of business/financial databases (Bloomberg, Datastream, Thomson One, Compustat) provides more flexibility for researchers to gather the panel data, however, these data sets are not primarily designed for OM empirical research. The data provided by them may not answering the research question in OM research. So the panel data may not be feasible in OM research (Ketokivi & McIntosh, 2017).

3.1.2.3.1.4. Generalized Method of Moments (GMM) Method

The generalized method of moments (GMM) method relies on transformations of existing variables rather than the use of external variables to address the endogeneity issues. And there are two common models, the System GMM and Difference GMM.

3.1.2.3.1.4.1. Difference GMM

Arellano and Bond (1991) proposed a GMM estimator that uses the lagged values of the endogenous regressors as instruments for the variables in the difference equations, which is known as the Difference GMM estimator (Yiu et al., 2020). Valid instruments, by definition, should be highly correlated with the variables to be instrumented but orthogonal to the error term in order to address the endogeneity concern (Wiengarten et al., 2017). Arellano–Bond test of autocorrelation

is used to check whether there are serial correlations in the idiosyncratic disturbances that will make the second lag invalid as instruments (Yiu et al., 2020). These requirements result in a set of “moment conditions” that enables the Difference GMM estimator to select the suitable lagged values as valid instruments for the difference equations. Blundell and Bond (1998) showed that the instruments used in the Difference GMM estimator could be weak if the autoregressive process becomes too persistent over time (Yiu et al., 2020).

3.1.2.3.1.4.1.1. Advantage of Difference GMM

Although the dynamic panel data model with the past value of the dependent variable included as the regressor can solve the endogeneity caused by the serial correlation of the dependent variable, it results in the “dynamic panel bias” or Nickell bias (Nickell, 1981) as the lagged dependent variable are correlated with the fixed effects in the error term by construction. After the transformation in Difference GMM, the constant term and the fixed effects in the error term are removed, so the concern of the Nickell bias is addressed (Yiu et al., 2020).

3.1.2.3.1.4.1.2. Disadvantage of Difference GMM

The lags used as the instruments in Difference GMM can be weak instruments for the difference equation if the dependent variable is close to a random walk (Blundell & Bond, 1998). The weak instruments correspond to weak identification of some or all of the unknown parameters. Weak identification leads to non-normal distributions, even in large samples, so that the Difference GMM inferences are misleading (Stock et al., 2002).

3.1.2.3.1.4.2. System GMM

Blundell and Bond (1998) developed a new GMM estimator with additional moment conditions in which the lagged differences of the endogenous regressors are used as instruments for the original-level equations, which is known as the System GMM estimator (Yiu et al., 2020). The System GMM estimation technique that relies on transformations of existing variables rather than the use of external variables to address the endogeneity.

The System GMM estimates a system of two equations simultaneously, i.e., the original-level equation and the transformed difference equation. As the fixed effects are still present in the error term of the original-level equations, an additional assumption is required in order to implement the System GMM estimator. Although the endogenous variables are correlated with the fixed effects in the error term by construction, it is assumed that the correlation is constant over time (i.e., time-invariant) (Arellano & Bover, 1995; Blundell & Bond, 1998). This assumption allows the System GMM estimator to develop differences as instruments to make them exogenous to the fixed effects in the error term and address the endogeneity concern (Roodman, 2009).

The System GMM estimator uses lagged differences as instruments for the original-level equations, in addition to the use of lagged levels as instruments for the transformed difference equations. The introduction of more instruments enables the System GMM estimator to address the concern of weak instruments in the Difference GMM estimator and improve the estimation efficiency dramatically (Roodman, 2009).

The lagged dependent variables are used as instruments to mitigate the dynamic endogeneity for the panel data and these instruments are defined as **internal instrument** (Lam et al., 2016). The

GMM model can mitigate endogeneity caused by different source, namely “unobserved heterogeneity, simultaneity and dynamic endogeneity” (Wintoki et al., 2012). In other words, GMM model can mitigate endogeneity caused by omitted variables, errors-in-variables and simultaneity (Lam et al., 2016; Yiu et al., 2020).

And System GMM estimator to have much better performance under finite sample properties in terms of bias and root mean squared error (rmse) than that of the difference GMM estimator (Bun & Windmeijer, 2010).

3.1.2.3.1.4.2.1. The Procedures to Implement System GMM Model

The System GMM model mitigate endogeneity by “internally transforming the data”–transformation refers to a statistical process where a variable's past value is subtracted from its present value (Roodman, 2009). The internal transformation process reduces the number of observations and thus improve the efficiency of the GMM model (Yiu et al., 2020). There are two kinds of internal transformation methods, which are first-difference transformation (one-step GMM) and second-order transformation (two-step GMM). And these two method can also be utilized as GMM estimators. There are limitations of the first-difference transformation (one-step GMM), such as the loss of too many observations if a variable's recent value is missing in this model. Therefore, to avoid potential data loss owing to the internal transformation problem with the first-step GMM, the use of a second order transformation (two-step GMM) is recommended (Yiu et al., 2020).

For the second-order transformation (two-step GMM), the ‘forward orthogonal deviations’ is utilized. It means that the two-step GMM model subtracts the average of all future available observations of a particular variable (Roodman, 2009). Because the two-step GMM model can avoid the data loss mentioned above, the two-step GMM model can provide more efficient and consistent estimates for the involved coefficients for a balanced panel dataset (Yiu et al., 2020). And the GMM model relies on internal instruments (lagged values, internal transformation) to address the different sources endogeneity, which is the main difference compared with the instrumental variable techniques that use exogeneous instruments.

Forth, the System GMM estimator addresses the dynamic panel bias directly by instrumenting the lagged dependent variables with variables uncorrelated with the fixed effects in the error term (Roodman, 2009). Fifth, the System GMM estimator is suitable for our data with unbalanced panels. the System GMM estimator appears to be one of “the most robust methodologies for unbalanced panels with endogenous variables”. Finally, although the System GMM estimator also employs the IV techniques to deal with the endogeneity issue, it does not rely on external variables that are outside the immediate dataset to construct instruments. Instead, the System GMM estimator constructs instruments internally with the transformation of existing variables, overcoming the difficulty in obtaining exogenous instruments externally (Roodman, 2009; Wintoki et al., 2012).

3.1.2.3.1.4.2.2. Post-Estimation Test of System GMM

When applying the System GMM model in research, two post-estimation tests need to be performed to determine that an appropriate econometric model is applied. The two tests are: (1)

the Sargan test; and (2) the Arellano-Bond test for first-order and second-order correlation (Yiu et al., 2020).

Sargan test determines whether the econometric model is valid or not, and whether the instruments are correctly specified or not. In other words, if the null hypothesis is rejected, the researcher needs to reconsider the model or the instruments used in the estimation process.

To examine the validity of a strong exogeneity assumption, the Arellano-Bond test for no autocorrelation (or no serial correlation) is used under the null hypothesis that the error terms of two different time periods are uncorrelated (Yiu et al., 2020).

3.1.2.3.1.4.2.3. Advantage And Disadvantage of System GMM

The System GMM mode has many advantages. It offers more efficient and consistent estimates for the coefficients as compared with other estimation techniques. It provides a consistent, asymptotically normal, and efficient estimator among all the estimators that only use the information provided by the moment conditions.

First, the System GMM estimator is suitable applicable to “small T, large N” panels (Roodman, 2009; Yiu et al., 2020). Second, as the System GMM estimator is one of “the most robust methodologies for unbalanced panels with endogenous variables” (Flannery & Hankins, 2013), it is appropriate for the research with some samples having more observations than the others and in which a number of endogenous variables are presented (Yiu et al., 2020). Third, the System GMM estimator enables us to address the Nickell bias by transforming the error term to remove the fixed effects (in difference equation) and by instrumenting the lagged financial returns with variables orthogonal to the fixed effects in the error term (in level equation) (Yiu et al., 2020). Forth, the System GMM estimator relies on the transformation of existing variables rather than the use of external variables to construct instruments (Roodman, 2009; Wintoki et al., 2012). This is an important advantage for the research with difficulty to obtain appropriate exogenous instruments externally (Yiu et al., 2020). Fifth, the System GMM can address the concern of weak instruments in the Difference GMM estimator and improve the estimation efficiency dramatically (Roodman, 2009), because the System GMM uses lagged differences as instruments for the original-level equations and lagged levels as instruments for the transformed difference equations. Sixth, System GMM estimator to have much better performance under finite sample properties in terms of bias and root mean squared error (rmse) than that of the difference GMM estimator (Bun & Windmeijer, 2010). Seventh, the System GMM model can mitigate endogeneity caused by different source, namely “unobserved heterogeneity, simultaneity and dynamic endogeneity” (Wintoki et al., 2012). In other words, GMM model can simultaneously mitigate endogeneity caused by omitted variables, errors-in-variables and simultaneity (Yiu et al., 2020). System GMM is convenient for estimating interesting extensions of the basic unobserved effects model, for example, models where unobserved heterogeneity interacts with observed covariates (Wooldridge, 2001; Yiu et al., 2020). And System GMM estimator has better performance than other methods, when it is applied to unobserved effects models when the explanatory variables are not strictly exogenous even after controlling for an unobserved effect. Besides, System GMM is well suited for obtaining efficient estimators that account for the serial correlation existing in the time-series data of the dependent variable. It can be applied to a model contains a lagged dependent variable along with an unobserved effect (Yiu et al., 2020).

Comparison with Instrumental Variable Techniques: For instrumental variable techniques, it is difficult to obtain such strictly exogenous instruments externally as pointed out by prior studies (Bardhan et al., 2013; Wintoki et al., 2012). In contrast, System GMM use internal instruments. Therefore, System GMM has several advantages. First, the System GMM controlled for different kinds of endogeneity by including lagged values of the dependent variable (previous value of the dependent variable) as an explanatory variable in the model. The System GMM is well suited for obtaining efficient estimators that account for the serial correlation. In contrast to 2SLS and 3SLS, it does not rely on external exogenous instruments, which in practice may be difficult to identify (Wintoki et al., 2012), but consists of a system of two sets of equations, each with its own internal instruments (Lam et al., 2016; Wiengarten et al., 2017). Thus, System GMM model is more useful when exogeneous instruments of good quality (i.e., valid and strong) are not available. Second, the System GMM estimator also can be used to generate standard errors that are robust to autocorrelation (in time series or long panel data) in the data (Yiu et al., 2020). This cannot be achieved by instrumental variable techniques, because the instrumental variable techniques do not allow the lagged value. Third, the System GMM model can be more efficient (i.e., smaller standard errors) if errors are considered to have arbitrary heteroscedasticity or intra-cluster correlation for over-identified equations. The optimal GMM estimator is asymptotically no less efficient than two-stage least squares under homoskedasticity, and GMM is generally better under heteroskedasticity. Meanwhile, the GMM estimator has similar performance with 2SLS for just-identified models, but better performance than 2SLS for over-identified models. Forth, the System GMM can be applied more often to unobserved effects models when the explanatory variables are not strictly exogenous even after controlling for an unobserved effect. For example, when the fixed effects are eliminate by differencing the observations of adjacent years, if the errors in the first-differenced equation are homoscedastic and serially uncorrelated, the 2SLS estimator is efficient; if not, a GMM estimator can improve upon two-stage least squares.

Comparison with AR and FE model: The AR and FE models are static panel data model. While, the System GMM is a dynamic panel data model. In other words, an important difference between FE model and GMM is that FE estimation uses ‘strict exogeneity’ assumptions resulting into a static FE model. For panel data, the past realization of dependent variables may also affect realization of dependent variables of the current year and the simultaneity could potentially violate the strict exogeneity assumptions. Under such situations, traditional FE and AR panel data static models will provide inconsistent and biased results (Wooldridge, 2012). Conversely, the System GMM estimation is more robust in dealing with these sources of endogeneity.

Comparison with dynamic panel data (DPD) model: Although the use of lagged dependent variable as the regression in the regression model can solve the endogeneity caused by the serial correlation of dependent variable, it can cause the “dynamic panel bias” or Nickell bias (Nickell, 1981) as the lagged value of dependent variables can be correlated with the fixed effects in the error term by construction (Yiu et al., 2020). This correlation makes the conventional ordinary least squares (OLS) estimation inconsistent (Roodman, 2009). Although a least squares dummy variables (LSDV) estimator can address this bias (Kiviet, 1995; Lam et al., 2016), it does not deal with the other endogeneity concerns such as unobservable heterogeneity and simultaneity arising from the other regressors included in the research (Lam et al., 2016; Roodman, 2009; Yiu et al., 2020). GMM model can simultaneously mitigate endogeneity caused by omitted variables, errors-

in-variables and simultaneity (Yiu et al., 2020). Under such situation, System GMM estimator should be used.

There are also several disadvantages of System GMM. First, the System GMM method ignores cross-sectional dependence (Roodman, 2009). Second, it assumes that the panel members have homogenous slope coefficients (homogenous panel) (Roodman, 2009).

3.1.2.3.1.4. Lagging independent variable

The lagging independent variable method can mitigate the endogeneity caused by simultaneity. It is a lagged endogenous regressor technique, which is very similar to the GMM model. A temporal separation through introducing a time lag between the independent and dependent variables can reduce this bias (Tsikriktsis, 2007). In this method, the dependent variable is measured in a time-lagged fashion and the past values of dependent variable is also included as the regressors in the regression model (Wiengarten et al., 2017). In this way, the endogeneity concern stemming from simultaneity effects can be significantly alleviated.

The advantage of using lagging independent variables is that it allows researchers to test for potential lagged effects based on the longitudinal data. In a longitudinal study, researchers can incorporate time lags between variables and to move a step closer toward understanding “cause and effect” in empirical operations management research (Tsikriktsis, 2007).

However, this approach also has limitation. For example, it is possible that the value of dependent variable at year t-2 can influence the value of the dependent variable in the year t-1, and these two value are still correlated with the value of the dependent variable in the year t. Under this situation, the lagged collaborative regressors are still endogenous. Besides, the models use lagged variables may have lower explanatory power than other models, such as the cross-sectional regression model (Tsikriktsis, 2007).

3.1.2.3.2. Longitudinal, quasi-experimental data

A quasi-experiment aims to evaluate interventions but that do not use randomization. Similar to randomized trials, quasi-experiments aim to demonstrate causality between an intervention and an outcome (DeHoratius & Raman, 2007; Dhanorkar & Muthulingam, 2020; Lam et al., 2022). The quasi-experiments allow a before-and-after type analysis, which enables consistent estimation of the effect of the intervention, given that subjects essentially act as their own controls in terms of stable observed and unobserved factors (Ketokivi & McIntosh, 2017). And the outcome is measured at multiple points before and after the intervention, and the objective to determine whether there is a discernible discontinuity in the time series at the intervention point (Ketokivi & McIntosh, 2017). And the difference-in-difference (DID) method, matching methods and synthetic control method can be used in the quasi-experiment design to make casual inference.

3.1.2.3.2.1. Difference-in-Difference Method

Difference-in-difference is a quasi-experimental design that using longitudinal data from treatment and control groups to obtain an appropriate counterfactual to estimate a causal effect. It is a method to estimate the effects of new policies. It consists of identifying a specific intervention or treatment. Then, the difference in outcomes after and before the intervention or treatment for groups affected

by the intervention to the same difference for unaffected groups should be compared (Bertrand et al., 2004).

When the treatment variable is time-varying and binary, one can apply the DID approach with panel data (Mithas et al., 2022). And to effectively apply DID analysis, the **parallel trend assumption** must be met (Fan et al., 2022; Lam et al., 2022). This means that the characteristics between treatment and control groups are the same overtime without the treatment in DID. An effective method to prove that this assumption is fulfilled is to visually assess whether the pre-treatment trends between the treated and control groups evolve in parallel (Mithas et al., 2022). And the parallel in the pre-trends can be tested in an event study analysis/relative time model, where the mean difference in the outcomes between the treated and control units is estimated period-by-period, with several periods before and after the treatment (Mithas et al., 2022). To ensure the pre-trends are parallel between the treatment and control groups, the matching methods, such as the propensity score matching(PSM) (Fan et al., 2022; Lam et al., 2022), and synthetic controls (C. J. Chen et al., 2022; Miao et al., 2023) can be applied. It has also been proposed that the smaller the time period tested, the more likely the parallel assumption is to hold. Violation of parallel trend assumption will lead to biased estimation of the causal effect (Mithas et al., 2022).

DID is an effective tool to apply when randomization on the individual level is not possible . This approach eliminates biases in post-intervention period comparisons between the treatment and control group that could be the result from permanent differences between those groups, as well as biases from comparisons over time in the treatment group that could be the result of trends due to other causes of the outcome. Therefore, it can control for the unobserved variables that bias estimates of causal effects, aided by longitudinal data, DID can mitigate endogeneity caused by omitted variables, errors-in-variables and simultaneity.

3.1.2.3.2.1.1. Assumptions of Difference-In-Difference

To effectively apply DID in research, four assumptions should be fulfilled. The assumptions are:

- (1) Exchangeability-Intervention unrelated to outcome at baseline (allocation of intervention between treatment group and control group was not determined by outcome)
- (2) Positivity-Treatment/intervention and control groups have Parallel Trends in outcome. And this assumption is also known as the Parallel Trend Assumption.
- (3) Stable Unit Treatment Value Assumption (SUTVA)-Composition of treatment and control groups is stable for repeated cross-sectional design.
- (4) SUTVA-No spillover effects

3.1.2.3.2.1.2. Advantage and Disadvantage of Difference-In-Difference Method

There are many advantages of DID. First, the result can be intuitive interpreted. The result obtained from DID can be easily interpreted by research based on intuition (Mithas et al., 2022). Second, the a relatively clean casual effect can be obtained using observational data if assumptions are met (Fan et al., 2022; Lam et al., 2022). Third, DID methods can use individual and group level data, which is very flexible (Li & Zhan, 2019). Forth, the control groups can start at different levels of the outcome, because DID focuses on change rather than absolute levels (Gong et al., 2023). Finally, DID accounts for change due to factors other than intervention (Fan et al., 2022; Lam et al., 2022). In other words, DID can control confounding variables in the research.

This method also has many limitations. First, DID require baseline data and a control group. The data before the intervention and the control group may not be available (Mithas et al., 2022). Second, if some key assumptions are not fulfilled, this method cannot be applied. If the allocation of the intervention is decided by baseline outcome, DID cannot be used. And if the control groups have different outcome trend with the treatment groups, the method cannot be applied (Fan et al., 2022; Lam et al., 2022). Meanwhile, if the composition of groups before and after the intervention are not stable, the method cannot be used (Wang & Overby, 2023).

3.1.2.3.2.2. Matching Method

Matching method is a quasi-experimental method in which the researcher uses statistical techniques to construct an artificial control group by matching each treated unit with a non-treated unit of similar characteristics. It is a method deal with the selection bias caused by the sample self-selection on observables. This method solve the selection bias by matching the observable features between the treatment group and the control group. Therefore, matching method can mitigate endogeneity caused by omitted variables. And it is useful for estimating the impact of a program or event for which it is not ethically or logistically feasible to randomize. There are three kinds of matching methods: (1) Direct Matching Method (2) Propensity Score Method (PSM) (Fan et al., 2022; H. Li et al., 2022; Qiu et al., 2022) (3) Coarsened Exact Matching (CEM) (Lo et al., 2022; Qiu et al., 2022).

3.1.2.3.2.2.1. Assumption of Matching Method

There are two assumptions for the matching method. The first assumption is the conditional independence, which states that after conditioning on a set of observed covariates, treatment assignment is independent of potential outcomes. The second assumption is the common support condition, which indicates the regions of stratification share enough members of the treatment and control groups. To effectively apply the matching method, the assumptions should be fulfilled.

3.1.2.3.2.2.2. Direct Matching Method

Direct matching method is a matching method that match each treated unit with a non-treated unit of based on similar observable features, directly (Lam et al., 2022; Liu et al., 2014; Lo et al., 2013, 2014; Wiengarten et al., 2019; Yeung et al., 2011). It is an ideal matching method because it ensures that the observable features of control group and treatment group are same. However, when the dimensions of features increase, it will be very difficult or even impossible to perform matching process directly based on all features. Then, the PSM can be used.

3.1.2.3.2.2.3. Propensity Score Method

Propensity Score Matching (PSM) is a matching method that computes that probability that a unit will enroll in the program. This probability is called the propensity score and is used to match units in the treatment group with unenrolled units of similar propensity scores (Fan et al., 2022; H. Li et al., 2022; Qiu et al., 2022). There are four stages in PSM, that are: (1) Estimation of propensity score (2) Selection of a matching algorithm (3) Checking for balance in the characteristics of the treatment and control groups (4) Estimating the program effect and interpreting the results (Fan et al., 2022; H. Li et al., 2022; Qiu et al., 2022).

Estimation of propensity score: The propensity score is also the probability of participation in the program for each unit in the sample. And the logit or probit regression method is used to obtain this score (Fan et al., 2022; H. Li et al., 2022; Qiu et al., 2022).

Select a matching algorithm: After the propensity scores are estimated, units in the treatment group are then matched with non-beneficiaries with similar propensity. The most commonly used matching algorithms in PSM are:

- (1) Nearest-neighbor matching: It is the one of the most commonly applied matching algorithm in OM empirical research (Dhanorkar & Muthulingam, 2020; Fan et al., 2022; H. Li et al., 2022). Each unit in the treatment group is matched with the n units in the control group with the closet propensity score, where n is the number of units that can be matched with the treated units. Units in control groups for which there are not unit in the treatment groups with a sufficiently similar score are removed from the sample. And the same is true for units in treatment groups for which there is not similar unit in control groups. A variation of nearest-neighbor matching matches multiple units in control groups to one single unit in treatment groups. If the unit in the control group can be used repeatedly, the bias decreases while the variance increases. And if the number of units in control groups that can be used to match with each treated unit increases, the bias increases and the variance decreases.
- (2) Radius matching: (i.e. ‘caliper’ matching): A maximum propensity score radius – a ‘caliper’ – is established, and all units in control groups within the given radius of a unit in treatment groups are matched to that treated unit. When the value of radius increases, the bias increases and the variance decreases.
- (3) Kernel matching: For each treated subject, a weighted average of the outcome of all untreated unit is derived. The weights are based on the distance of the propensity score of untreated units to that of the treated subject’s, with the highest weight given to those with scores closest to the treated unit. When the bandwidth increases, the bias increases and the variance decreases.

The selection of matching algorithm should be based on the trade-off between the bias and the variance.

3.1.2.3.2.2.3.1. PSM in OM Research

In OM empirical, PSM is commonly used to ensure that the treatment and control firms are highly similar (Dhanorkar & Muthulingam, 2020; Petryk et al., 2022). Qiu et al. (2022) use PSM approach aims to calculate the probability (i.e., propensity score) of being treated/accept treatment for their treatment and control firms. They use the nearest neighborhood approach to match each treatment firm to a control firm with the closest propensity in the same industry (two-digit SIC code) in the research. By the nearest neighborhood approach, one control firm is matched to multiple treatment firms, which may cause a double-counting issue. To solve this issue, they remove the treatment firms that could not be matched with any control firms. They calculate the propensity score for each treatment and control firm based on the estimation coefficients- Pseudo-R-squared. And the range of variance inflation factor is between 1.00 and 1.28, indicating that multicollinearity is not a serious concern. And they perform paired t-test for samples after matching to ensure no selection bias in the post matching samples. They conduct paired t-tests on all the independent variables (except industry dummies) used in the probit model. The test results indicated no significant difference between treatment and control firms on these variables ($p > .1$) after PSM. Thus, they concluded that the statistics of treatment and control firms are highly similar.

3.1.2.3.2.3.2. Advantages and Disadvantages of PSM

There are some advantages of PSM method. First, PSM is an approximate matching approach (Qiu et al., 2022). Second, PSM can solve problems caused by selection bias (Fan et al., 2022; H. Li et al., 2022; Qiu et al., 2022) and help researchers to make causal inference. Using probit or logit regressions, the PSM method pairs every treatment sample with a sample from the control group to create a quasi-control group and randomizes the data effectively. The matching method helps researchers to compare and contrast two statistically twinned firms to examine the treatment effect bias (Fan et al., 2022; H. Li et al., 2022; Qiu et al., 2022).

There are also some disadvantages of PSM method. First, it can only solve endogeneity caused by omitted-variable endogeneity caused by observable confounding events, however, PSM cannot solve the endogeneity caused by simultaneity and unobservable confounding events (Qiu et al., 2022). Second, to build quasi-control group required by PSM method, a relatively large secondary dataset of control subjects is needed, which may be not available in specific research. Third, PSM method is based on an overall score between the samples and control, so even the total propensity scores are similar, the IVs between the two may not be the same and sometime significantly different (Lo et al., 2022).

3.1.2.3.2.2.4. Coarsened Exact Matching

Coarsened exact matching (CEM) is another matching method, wherein continuous and ordinal characteristics are categorized and values of categorical variables are collapsed (i.e., coarsened). CEM is faster, easier to use, requires fewer assumptions, robust to measurement error, and strictly bounds both the degree of model dependence and the average treatment–effect estimation error through ex ante user choice (Qiu et al., 2022). To implement the CEM method, we need to first coarsen our data, determine an exact match on these coarsened data, and then conduct our analysis on the un-coarsened, matched data.

3.1.2.3.2.2.4.1 Steps to Perform Coarsened Exact Matching

To implement the CEM method, we need to first coarsen our data, determine an exact match on these coarsened data, and then conduct our analysis on the un-coarsened, matched data (Qiu et al., 2022). Steps to conduct CEM is described as follows (Blackwell et al., 2009; Lo et al., 2022):

Step 1: Begin with the covariates samples or observations X and make a copy of them, which we denote X^* .

Step 2: Coarsen X^* according to user-defined cut-points, or CEM's automatic binning algorithm.

Step 3: Create one stratum per unique observation of X^* and place each observation in a stratum.

Step 4: Assign these strata to the original data, X and drop any observation whose stratum does not contain at least one treated and one control unit.

3.2.1.3.2.2.4.2. Advantage of Coarsened Exact Matching

There are some advantages of CEM method. First, CEM is faster, easier to use, requires fewer assumptions, robust to measurement error, and strictly bounds both the degree of model dependence and the average treatment–effect estimation error through ex ante user choice (Qiu et al., 2022). Second, CEM can offer a better way to control the differences between the IVs for sample and control groups in the model (Lo et al., 2022). Since PSM matching is based on an overall score between the samples and control, so even the total propensity scores are similar, the

IVs between the two may not be the same and sometime significantly different. CEM method can solve this problem in PSM.

The disadvantage of CEM method is that it does not allow for the estimation of average treatment effect in the matched sample because the matched units in each treatment group will not resemble the overall sample (Lo et al., 2022; Qiu et al., 2022).

3.2.1.3.2.4. Limitations of Matching Method

There are two major limitations. First, extensive datasets are required to properly match units and they rely on broad assumptions that are difficult to prove. In research, there may not be enough data to perform the matching method. Meanwhile, the datasets should have detailed information on baseline characteristics which is not always available. Second, the validity of matching methods relies on the assumption that there are no systematic differences in unobserved features between the treatment units and the matched units in control groups (Qiu et al., 2022). It is difficult to prove this assumption correct, making matching methods a less robust approach than other methods.

3.1.2.3.2.3. Synthetic Control Method

Synthetic control method (SCM) is a generalization of the difference-in-differences model. It is designed for comparative case studies in small and moderately sized samples as a transparent and data-driven way of constructing a synthetic control group for use in comparing the outcome of interest in an affected unit with the outcome of that control group without the direct exposure of that control group to the intervention of interest (Abadie, 2021). In short, SCM can evaluate the impact of a particular intervention (say, an event) on particular outcomes of interest. Therefore, SCM can be applied in the quasi-experiment design. It can solve endogeneity caused by selection bias.

The central principle of the SCM is that a linear combination of control entities provides a better comparison for a treated entity than any single control (C. J. Chen et al., 2022). An synthetic control unit is constructed for each treated entity by assigning weights to the available control entities. The weights are derived using pre-policy intervention data and assigned to minimize the difference between the treated entity and the associated SC with respect to both the outcome variable and observed covariates (J. Chen et al., 2023).

SCM relies on pretreatment outcome matching (Imbens & Wooldridge, 2009) and uses pre-intervention trends in outcomes and auxiliary covariates to simulate a counterfactual outcome scenario reflective of the hypothetical absence of the intervention itself. The central idea of SCM is to construct the outcome of interest from an artificial control group of units that have never experienced the intervention, but which have similar outcome dynamics and covariate levels as those of the treated unit in the pre-intervention period (Gilchrist et al., 2023). This is achieved by constructing a weighted average of control units that match the treated unit's outcome and auxiliary covariate values prior to the intervention. If the past outcomes and covariates are reasonably well-balanced between the treated unit and its control group, the difference in the outcome between the treated unit and its artificial (i.e., synthetic) control group plausibly captures the impact of the policy or intervention in question. In this respect, the idea of the SCM is to compute the gap in the given outcome of interest for the treated unit by assuming that the intervention or policy of interest had never happened thereby exploiting pre-intervention trends in

the outcomes and covariates between the treated unit and its control group (McClelland & Gault, 2017).

The treatment scope should satisfy three specific criteria. First, the shock assigned to the affected unit should not be easily anticipated by prior economic conditions. If these conditions predict the timing of the treatment, the conditional independence assumption on the treatment of interest most likely fails and masks the estimated counterfactual scenario with the pre-shock outcome dynamics that are not necessarily attributed to the treatment of interest. Second, as a treatment itself, it provides a plausible shock that mimics the characteristics of the random assignment. Third, the general thrust of the treatment selection is its uniqueness.

3.1.2.3.2.3.1. Advantage of SCM

There are several advantages of SCM. First, SCM is superior than matching methods because it not only matches on covariates (i.e., pre-treatment variables), but also outcomes. Second, SCM can also be used under the Bayesian framework where we do not have to impose any restrictive priori. Data-driven processes of constructing the synthetic control group are based on selecting weights of past outcomes and auxiliary covariates through the diagonal matrix comparison that determines which comparison units' outcome process characteristics are within the convex hull of the affected unit to best reproduce the outcome trajectory for the treated unit in the hypothetical absence of the intervention (Gilchrist et al., 2023). Third, it provides a sound basis for identifying structural and other economic changes that may then be the focus of historical study. Forth, SCM can yield a plausible assessment of the impact of the policy or intervention in question provided that the imbalance in the pre-intervention outcomes and auxiliary covariates is minimized. Fifth, From the selection criteria, researchers can understand the relative importance of each candidate. Finally, observable similarity between control and treatment cases is maximized (C. J. Chen et al., 2022).

3.1.2.3.2.3.1.1. Advantage of SCM Over Difference-in-Differences

- (1) Maximization of the observable similarity between control and treatment (maybe also unobservables).
- (2) Can also be used in cases where no untreated case with similar on matching dimensions with treated cases.
- (3) Objective selection of controls.

3.1.2.3.2.3.2. Disadvantage of SCM

The disadvantage of SCM is that it is difficult to argue for the weights used in the research can create the “synthetic control” (Abadie, 2021).

3.2.1.4. Problem with Longitudinal Research Designs

The general problems that are common for different research methods used in the longitudinal research design are discussed in this part. First, it is difficult to obtain panel data. OM researchers often have to build their datasets from scratch. The extant commercial databases are mainly designed for strategy or economics research and may not satisfy the demands of OM research. There are no commercial datasets that contain, such as longitudinal factory-level information about technology and productivity (Ketokivi & McIntosh, 2017). Second, since the intervention or exogenous shocks happened in the past, the OM researcher should be lucky enough to find a

meaningful exogenous shock that has an impact of theoretical interest (Ketokivi & McIntosh, 2017).

3.1.2.5. Instrument Free Approaches: Latent Instrumental Variables

The challenges related to identifying valid instruments have resulted in the emergence of instrument-free techniques. The Latent Instrumental Variables (LIV) method is one of the instrument free approaches. In LIV method, the variations in the endogenous independent variable are separated into exogenous (approximated by a latent discrete variable) and endogenous parts (Zhang, 2023). The exogenous part is uncorrelated with the disturbance term, while the endogenous part is probably correlated with the disturbance term (Ketokivi & McIntosh, 2017). And the exogenous part will be included in the regression model to mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

There are many advantages of this method. First, this method employs a latent variable model to account for regressor-error dependencies, and addresses the issues of instrument availability, weakness, and validity, because LIV estimator belongs to the family of thrifty instruments estimators that do not require observed instruments (Ebbes et al., 2009). Second, LIV estimator, as a maximum likelihood method, can achieve unbiased estimation. And In the LIV solution, a latent variable model is used to separate the endogenous covariate into systematic parts, whereby one part is uncorrelated with the error term and the other part is possibly correlated with the error term. This allows to achieve an unbiased estimate of the effect of an endogenous covariate on the dependent variable. The disadvantage of this method is that it requires the assumption that all regressors and the latent change components are uncorrelated with the disturbance term, which can be difficult to satisfy in specific research (Ketokivi & McIntosh, 2017; Zhang, 2023).

3.1.2.6. Instrument Free Approaches: Identification through Heteroskedasticity

The Identification through Heteroskedasticity (IH) estimator is also an instrument free approach. In this method, instruments are obtained in a similar manner to Higher Moments (HM) approach, which means that instruments are built based on available data in general regressor-error dependencies models. But the information of heteroscedasticity is required by IH method (Roberto, 2003). This IH method can be used to mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

Since the IH approach is a method-of-moments estimators, one limitation of this approach is that it often obtain biased estimation from the model. Because the method of moments is fairly simple and yields consistent estimators under very weak assumptions, though these estimators are often biased.

3.2. Solutions at Statistical Analysis Stage

The statistical analysis stage also called as post-design stage. There are several methods that can be used to mitigate the endogeneity at this stage.

3.2.1. Instrument Free Approaches: Higher Moments

The Higher Moments (HM) approach is also one of the instrument free approaches. In this approach, the instruments are built based on available data in general regressor-error dependencies models, and can be used together with or in the absence of traditional instruments (Erickson &

Whited, 2002; Holzmeister et al., 2020). The HM approach can mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

Since the HM approach is a method-of-moments estimators, one limitation of this approach is that it often obtain biased estimation from the model. Because the method of moments is fairly simple and yields consistent estimators under very weak assumptions, though these estimators are often biased.

3.2.2. Instrument Free Approaches: Copulas

The joint estimation with copulas is another instrument free method to deal with endogeneity issues. A copula is a function that ‘couples’ multivariate distributions to their one-dimensional marginal distribution function and captures the correlation between the independent variable and the disturbance term (Guo et al., 2021; Ketokivi & McIntosh, 2017). If the correlation is properly handled, this method can mitigate the endogeneity caused by omitted variables, errors-in-variables and simultaneity.

The likelihood of the structural equation of interest can be maximized by using a density estimation method in the model of the joint distribution of the endogenous variable and the disturbance term. This is achieved by copulas, where the functions that “couple” multivariate distributions to their one-dimensional marginal distribution functions and capture the correlation between the variable and the error (Ketokivi & McIntosh, 2017).

3.2.3. Heckman Two-Step Procedure

Heckman (1979) two-step procedure solves endogeneity caused by selection bias. The selection bias will occur when samples to be non-randomly selected by researchers. The non-random sample is subject to the selection bias that will cause the endogeneity issues.

There are two steps in this method. The first step is to run a probit regression (selection equation) to model the conditional distribution of the treatment with a set of covariates that captures the relevant features (Ketokivi & McIntosh, 2017; Wiengarten et al., 2019) Ketokivi & McIntosh, 2017. And the relevance of the selected set of covariates needs to be theoretically justified. Then, the propensity scores are estimated, which needs to be repeated for each independent variable of the model. The second step is to correct the sample self-selection by integrating the Inverse Mills Ratio (IMR) (H. Li et al., 2022; Qiu et al., 2022; Wiengarten et al., 2019), defined as the ratio of the standard normal density divided by the standard normal cumulative distribution function, into the final regression models before testing hypotheses (H. Li et al., 2022; Qiu et al., 2022; Wiengarten et al., 2019).

This method also has limitations. First, at least one variable with a non-zero coefficient in the selection equation in the first step should not be included in the final equation in the second step (H. Li et al., 2022; Qiu et al., 2022; Wiengarten et al., 2019). This variable is used as an instruments, which may be difficult to find. Second, a dummy variable only with two possible value of 0 and 1 is used to represent the probability of selection. In the first step of the model, a probit regression model is used to predict the probability of selection, so the predictor need to be a dummy variable that can take two discrete value of 0 and 1. To solve address this problem, one solution is to recode the predictor variable into a dummy one (Ketokivi & McIntosh, 2017). Another possible solution

is to use the two-step approach for selectivity-bias correction with a continuous choice variable (Garen, 1984). In method, first, regressing the independent variable against a set of factors that influence the independent variable. The output of this regression model is the choice variable, i.e. the probability of selection. In this procedure, the researchers generally include all control variables, independent constructs, and moderators except for the main dependent variable(s) into the correction regression model to predict the choice variable (H. Li et al., 2022; Qiu et al., 2022; Wiengarten et al., 2019). In the second step, the predicted errors from this correction regression equation should be added in to the second-stage performance equation (H. Li et al., 2022; Qiu et al., 2022; Wiengarten et al., 2019).

Second, not all factors that contribute to the occurrence of the selection bias can be measured in the selection equation. That is the omitted variables can still exist, causing the endogeneity issues. Therefore, other remedies may need to be used complementary to this method, such as PSM (Qiu et al., 2022).

4. Procedure to Solve Endogeneity

There are possible theoretical procedures for researcher to follow to handle the endogeneity. The detailed steps are showed as follows:

1. Where in the model the problem is-Identification of the endogeneity.

First, the endogeneity should be identified from the theory and extant literature, because endogeneity is based not on statistical theory but on substantive, theoretical considerations, and that is why evaluation of endogeneity is ultimately a judgment call (Lu et al., 2018). Researchers should list all the potential endogeneity concerns implied by the model, and then try to determine how serious a threat each of these pose to empirical analysis, from the theoretical aspect (Ketokivi & McIntosh, 2017). For example, endogeneity caused by simultaneity can be identified based on theoretical arguments or extant literature.

Second, endogeneity tests, i.e., statistical tests (e.g., Durbin–Wu–Hausman test), can be performed to identify the endogeneity. If the literature and/or theory suggest that endogeneity may be a concern, endogeneity tests should be performed to examine the extent of endogeneity.

2. What could be causing it-Identification the reason for the endogeneity

The sources of the identified should also be identified from theoretical aspect, first.

3. What can be done about it.

The first step in tackling endogeneity is to specify a theoretically sound model (Ketokivi & McIntosh, 2017; Lu et al., 2018). Researchers should build a theoretically grounded model based on the literature when possible and consult experts with extensive knowledge of the research context to gauge the extent of endogeneity. The endogeneity caused by omitted variables can be controlled by including control variables that are both theoretically and empiricall relevant to the theretical model in research. Meanwhile, endogeneity caused by simultaneity can be identified based on theoretical arguments or extant literature. And the endogeneity caused by errors-in-variables can be controlled by using appropriate and accurate measures according to the extant research.

After this, researchers can rely on empirical techniques to address endogeneity

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